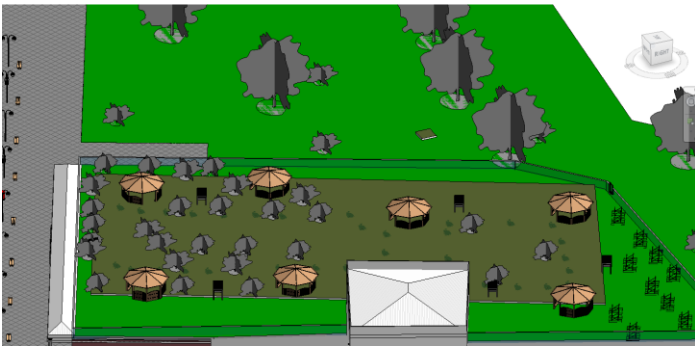




KA 7062

ASSESSMENT

Report on Northumberland Street (NLDS) Facility

W23005357

Digital Consultancy Report on
Northumberland Street (NLDS) Facility

1.0 Executive Summary

Green roofing is an innovative and sustainable strategy to enhance thermal insulation of roofs for improved energy performance of building envelopes, additionally offsetting embodied carbon in both existing buildings materials and the retrofitting process. Aligned with the objective of minimal disruptions to everyday operations of the NLDS facility, the exterior nature of the Green Roofs significantly reduces the risk of interferences.

This report encourages the adoption of Building Information Modelling (BIM) as an alternative to traditional Information Management (IM) practices on this retrofit project. It elucidates the intrinsic value of BIM adoption, delineates the specific applicability of BIM on the project, and provides a detailed process for implementation, including steps and the requisite information requirement. It focuses on the improved offerings of BIM in Energy Analysis and Cost Estimation during the design authoring process, scoping the role of BIM, implementation challenges and comparatively analyzing the limitation of Traditional IM.

Recognizing the significance of a holistic approach to the life cycle of a facility, this report explores the application of research-driven methodologies in the management of the NLDS facility. Particularly, it focuses on Predictive Maintenance (PM) and Energy Management (EM), leveraging proposed implementation frameworks, while identifying associated challenges.

Summarily, this report emphasizes the embedded potential of BIM in supporting retrofit projects through apt information management practices, specifically exploring energy analysis and cost estimation during design authoring and predictive maintenance and Energy management in Facility Management (FM).

2.0 BIM Improvement

Retrofitting buildings involves the alteration of existing systems, including the addition of innovative technology or features aimed at improving certain aspects of the building to achieve goals such as improved property value and energy efficiency. These endeavors face challenges related to information management and availability, compelling the adoption of efficient means of information definition, management and exchange, necessitating divergence from traditional IM practices (Silva et al., 2024). Information Management is a crucial aspect of retrofit projects, initiating a cascading effect, which influences the life cycle of a built asset. Herrera et al. (2021) denotes the effects of traditional IM on communication, buttressing its negative impact on collaboration, problem solving and information flow on project success.

The inherent complexity of retrofits, exacerbated by traditional IM hinders the ability to mine construction information for precise estimations of energy requirement and cost, engendering exaggeration. According to Langtry et al. (2024), insights generated from structured information support decision making by reducing uncertainties in the design of building energy systems. Additionally, the siloed and sequential nature of traditional IM encourages fragmentation and adversarial behaviors, which could result in exorbitant legal proceedings (Mohammad et al., 2023).

Eastman (2018) considers poor communication and documentation as the principal catalyst of poor estimates, noting that BIM through its computable and accurate nature improves quantity take off and cost estimations. He adds that BIM enhances the speed and reliability of cost estimations and encourages the use of BIM early in the project to limit risk and over exaggeration of project cost due to contingencies added to cater for overruns. Figure 1 adapted from Eastman (2018 p.163) shows a range of contingencies traditionally applied to cost estimates and reliability improvement offered by BIM-based estimation.

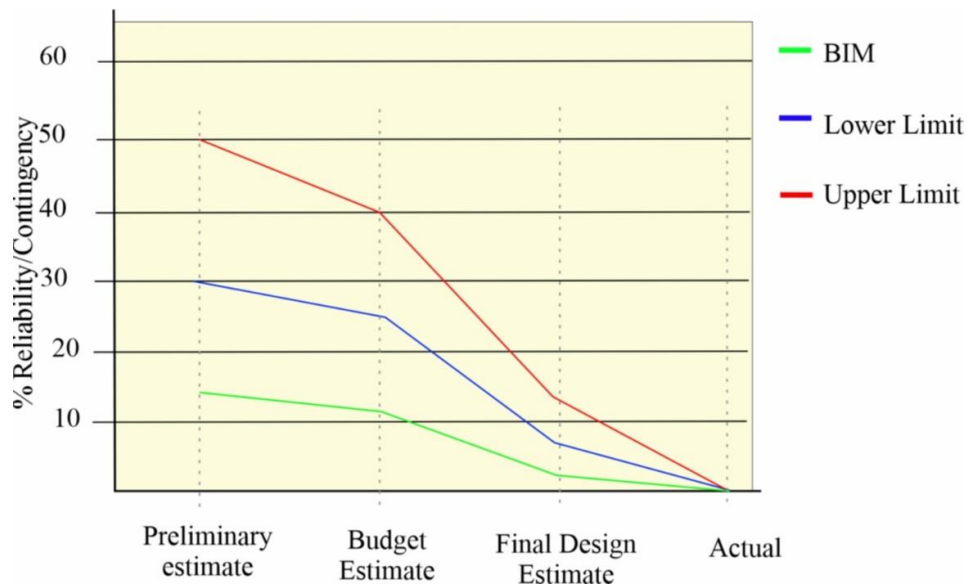


Figure 1 Effect of Information Management on Cost Contingencies

BIM is a structured process for model development and management aimed at accelerating the digitalization of the built environment (Pan & Zhang, 2021). It facilitates a wide range of analysis across the entire lifecycle of a retrofit project (Messner, 2023), from Planning and Design - pre-retrofit survey, auditing and performance assessment, option identification, through construction and management - implementation, validation and verification. The NLDS project will benefit from the implementation of BIM in energy analysis and cost estimations, which, through effective information management, optimizes resource utilization.

2.1 Energy Analysis for Design Authoring

Energy analysis extends the usability of BIM in the 6th dimension. It enables an environmental impact analysis of the existing condition of the NLDS facility and the effects of the proposed Green Roof to inform the iterative design process aimed at performance optimization. The predominant objective of this BIM application is to assess the facilities compliance to building standards and facilitate the optimization of the proposed design to positively influence the life-cycle cost and sustainability rating of the facility (Messner, 2023). According to Langtry et al. (2024), the UK built environment consumes 59% of generated electrical energy and is responsible for 23% of total greenhouse gas emissions. The proposed Green Roof helps reduce the impact of NLDS facility in this regard by serving as both an insulation layer and a decarbonizing agent.

According to the (Chartered Institute of Building Service Engineers, 2012), energy efficient buildings optimize the use of energy, minimizing its consumption thereof, whilst providing occupants with the required indoor environment and services in a cost effective and sustainable manner. The (Chartered Institute of Building Service Engineers, 2012) guide F table 20.1 provides an Energy Use Intensity (EUI) benchmark 270kWh.m^{-2} for existing retail stores and recommends the use of Dynamic Thermal Modelling (DTM) tools such as EnergyPlus for estimation of a building's energy requirement using Building Energy Models (BEM) in BIM. Hmidah et al. (2023) adds that the use of BIM leads to a 30% reduction in energy consumption on retrofit projects.

Before BIM, energy analysis required manual calculations using traditional construction information and simplified models, which did not consider the complexities of real-world situations. It was typically conducted post-construction making change implementation arduously exorbitant. (Eastman, 2018 p. 21) highlights the importance of BIM in decision-making based on its ability to develop schematic and detailed models, which allows

for careful evaluation of design proposals to evaluate its adherence to desired goals. According to Messner (2023), the use of BIM for energy analysis improves the degree of accuracy of energy predictions by auto-determining building information from BIM models (BIModel). It draws on embedded metadata to generate a detailed energy model of the NLDS facility to determine the environmental impact of the design, which informs the design optimization process and cross-reference it with sustainability standards to bring the facility a step closer to achieving the desired efficiency rating.

BIM-based technologies such as scan-to-BIM enhances the pre-retrofit survey process and involves the translation of laser-scanned point-cloud digital models into BIM platforms for BIModel development, giving accurate information on the present state of the facility. This is highly beneficial to the NLDS facility and similar retrofit projects where the as-built designs are inaccessible. In the performance assessment and option identification phase, BIM energy models enable the simulation of the building's energy performance to identify areas for improvement and facilitate the iterative design process of design alternatives, ensuring synchronous compatibility across the project in reference to the existing building, retrofit goals and objectives. Additionally, it provides information on energy performance and material selection for sustainability certifications.

The required resources for energy analysis of the NLDS facility includes, a team with knowledge of building energy systems, design and standards, a building Energy simulation and Analysis Software, a well-adjusted 3D BIModel of the facility with material properties including information on their thermal properties, local weather data and energy efficiency standards (Messner, 2023). The process map in figure 2 outlines the various information and processes deployed in energy analysis of the NLDS facility.

2.2 Cost Estimation for Design Authoring

The growing expectations for efficient project delivery and the adoption of lean methodologies have necessitated the development of integrated systems for cost estimation, monitoring and control (Pishdad & Onungwa, 2024). BIM enables the generation of precise quantify take-offs and estimates from BIModels though a project's life cycle and allows for dynamic adjustments of estimates in response to design modifications. This dynamic adjustment capability negates excessive budget overruns due to design adjustments, proving vital in the preliminary stages of project design (Messner, 2023).

Traditional cost estimation practices in the construction industry often rely heavily on manual methods and subjective expertise of individual actors. Estimators typically rely on manual measurement of quantities from blueprints, historic data and standard cost books to make educated predictions of projects cost. Unlike in traditional estimation approaches where material quantities are extracted from 2D drawings and manually manipulated to establish cost, BIM based cost estimation takes advantage of data rich 3D parametric models, drawing on embedded geometric data for quantity determination (Fazeli et al., 2021). This helps alleviate the associated redundancies of traditional estimation such as manual data entry, inconsistent data, stifled collaboration and frequent recalculations when changes occur, which are not just error-prone but adversely affect time and cost.

According to Hardin and McCool (2015 p. 4), the core value of BIM is its ability to extend the usability of BIModel information by making it relevant for other related workflows and processes including, quantification and estimation. The data population of 3D elements in the NLDS facility BIModel will enable its use in cost estimation. When BIM is parametrically managed on a project, low-level corrections are automatically made during design changes and updates, reducing the need for users to manage design changes, which enhances coordination and alignment amongst the project team and across project information (Eastman, 2018 p. 21).

This entails that in conjunction with the ability to generate Quantity Take-off (QTO), changes to geometry in the BIModel will automatically reflect on the QTO, ensuring accurate communication of quantities amongst project stakeholders, which enhances collaboration, by ensuring that suppliers are always equipped with up-to-date information, including full specification on the required components.

According to Pishdad and Onungwa (2024), cost-loaded BIM, manual data mapping and linked models are the routine information exchange process between BIM-based design authoring and estimation tools. This necessitates establishing appropriate BIM processes for the implementation of cost estimation depending on the choice of information exchange. The NLDS facility will determine the cost estimate for the proposed Green Roof using Cost-loaded BIM with cost attributes assigned to 3D geometric elements when created in Revit. The challenges of this method revolve around the need for continuous cost updates to cater for market fluctuations. The required resources for cost estimation of the NLDS facility includes a team with related competencies, a model-based estimation software (such as BEXEL), design-authoring platform and a precisely built design model with cost (Messner, 2023). The process map in figure 2 outlines the various information requirement exchanges and workflow required for cost estimation of the green roof of the NLDS facility.

2.3 Final Overview of Improvement

The figure below is a two-level Business Process Modelling Notation based process map deployed in the implementation of the detailed BIM Improvement as recommended in Messner (2023).

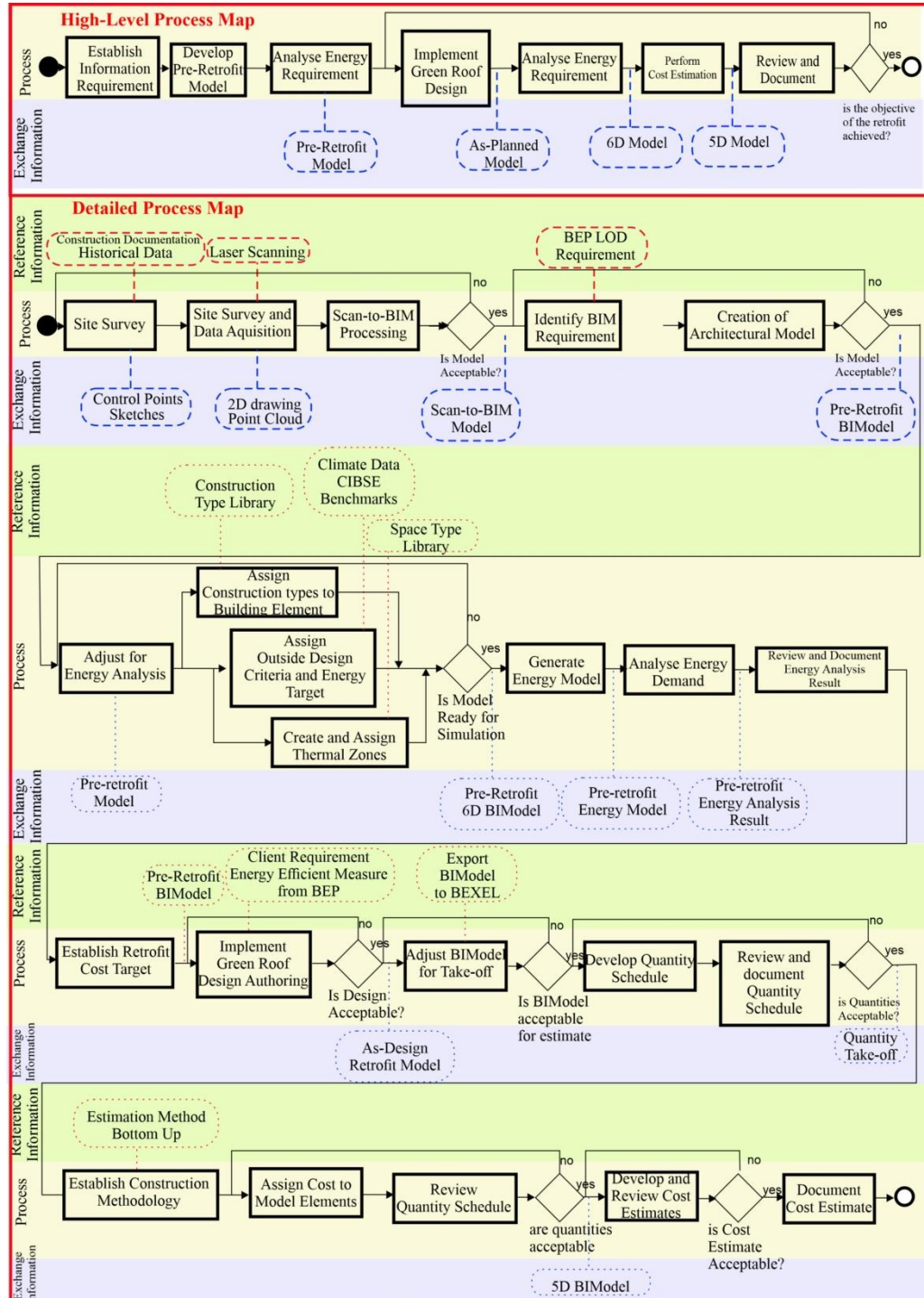


Figure 2: Implementation Process Map

The application of BIM for Energy Analysis and Cost Estimation informs the design authoring process of the green roof system for the NLDS facility. It enabled the precise analytical comparison of the impact of the retrofit to the execution cost for optimization along with the advantages mentioned in section 2.1 and 2.2.

The first step in this process involves establishing information requirements prior to drawing on historical data and construction documentation to establish control points for laser scanning to produce point clouds for processing and development of the As-is pre-retrofit BIModel in a scan-to-BIM process (see figure 3). Traditionally, this relies on manual measurement and the use of surveying equipment in a time consuming, tedious and error-prone process.

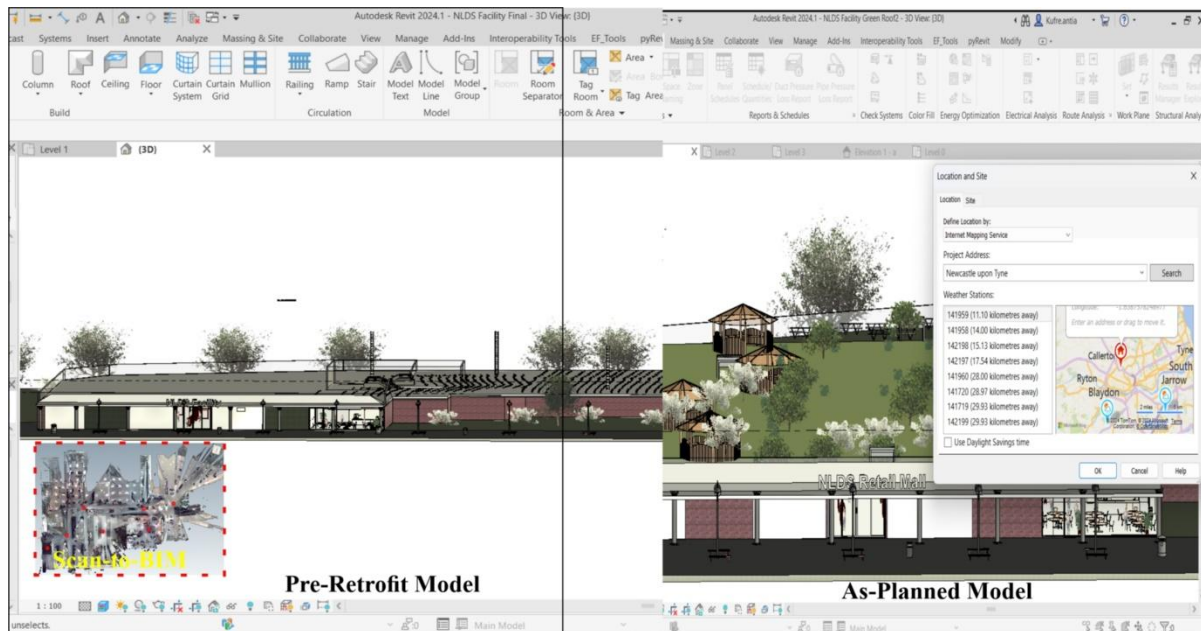


Figure 3: Pre-Retrofit and As-Planned Model

In contrast to traditional 2D CAD which lacks the capacity to embed data into building elements, the pre-retrofit and As-planned BIModel allows for energy analysis through the

population of elements with construction type data including material properties such as thermal conductivity and performance characteristics including thermal resistance and transmittance, air permeability and heat gain coefficient. Additionally, the process involves the integration of information on outside design condition by setting the BIModel coordinates and linking of weather station data to allow the model draw on information regarding solar positioning and climate for enhanced accuracy. Thermal zones are defined for precise load calculations as different areas of a building have varying heating and cooling requirements based on use, occupancy and exposure, prior to generating energy model and analysing results for documentation (see Figure 4).

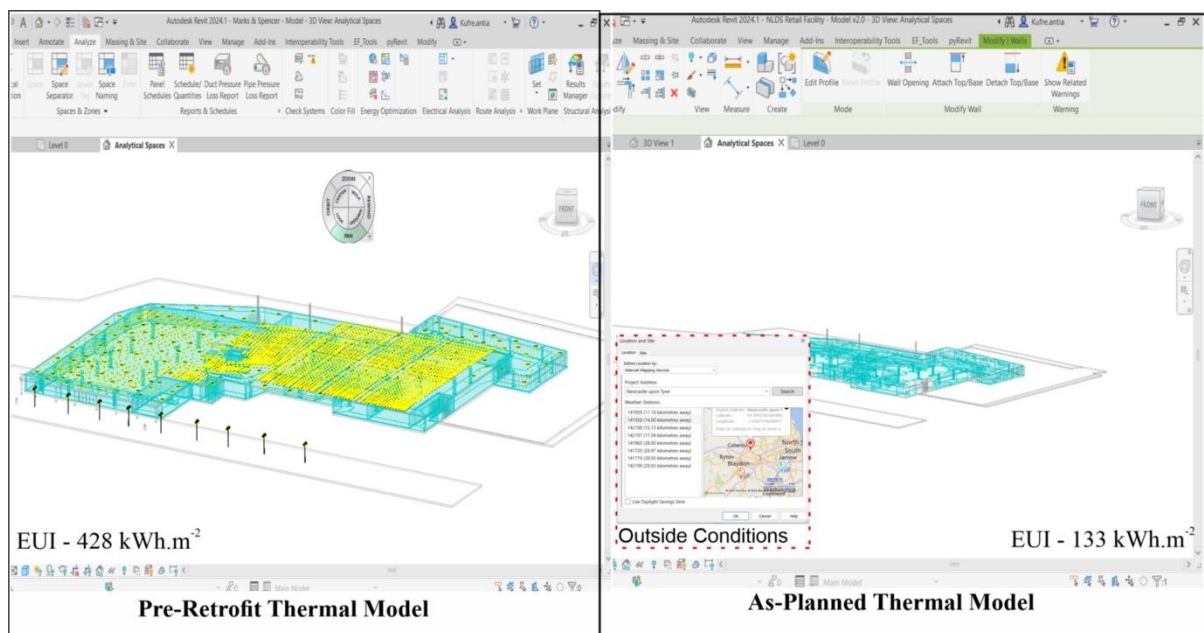


Figure 4 Pre-Retrofit and As-planned thermal Models

With established pre-retrofit conditions, design authoring begins the next phase, moving sequentially through concept, schematic and construction design phases. The Green Roof is designed adhering to documented client requirements and upon design acceptance is transferred and exported as the As-planned design for cost estimation with element properties

set to convert to include quantities information. This involves adjusting the BIModel for take-off by mapping work sets and ensuring that it meets the required LOD, prior to generating and reviewing the quantity schedule for documentation (see Figure 5).

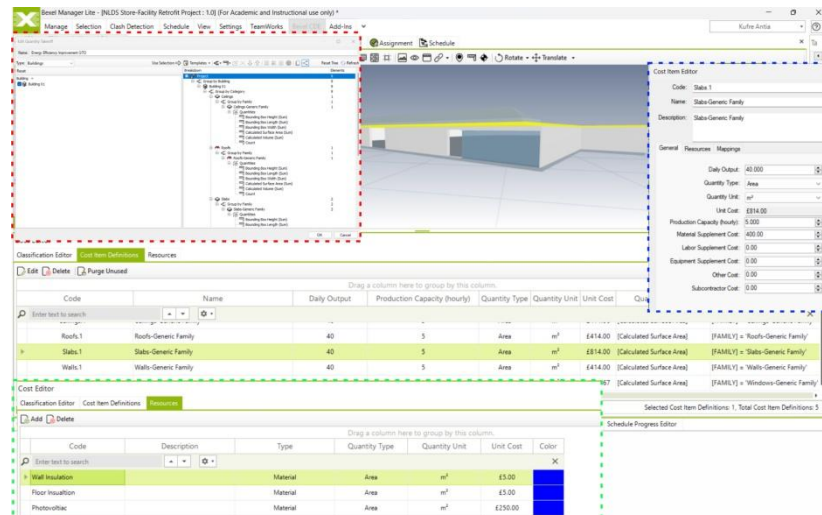


Figure 5: Adjusting BIModel for Quantity Take off and Estimation in BEXEL

Using Bottom-up estimation method and establishing construction methodology, project cost parameters and resources are created and linked to 3D elements in the model. Develop, review and document cost estimate at the end of every iteration for referencing (See Figure 5).

Code	Name	Quantity	Unit	Unit Cost	Material Cost	Labor Cost	Equipment Cost	Base Construction Cost
1	Roofs-1	26,002.240	m²	£414.00	£10,764,927.48	£24,699,407.05	£305,865.34	£130,011.20
2	Stairs-1	40	m²	£414.00	£16,560.00	£3,718.80	£46.48	£20,325.28
3	Walls-1	30,175.498	m²	£414.00	£12,512,757.41	£24,699,407.05	£305,865.34	£130,011.20

Figure 6: Cost Estimate For Green Roof

3.0 BIM for Facility Management

Facility Management (FM) is a multifaceted discipline encompassing the management of built assets to ensure optimal performance and comfort for occupants through services in maintenance and repair, work place experience, space utilization, technology integration, vendor, emergency, sustainability and energy management aimed at cost efficiency, enhanced productivity, risk mitigation and sustainability (Manchana, 2022). Effective FM enables organisational success by supporting operational continuity through the optimization of building systems, adept maintenance strategies and compliance with sustainability regulation thus reducing cost and enriching the overall quality of the organisation and environment. Post-construction operation accounts for approximately 85% of the total lifecycle cost of a facility with an estimated USD20 billion lost annually during FM phase in the United States (Lewis et al., 2010, in Korpela et al., 2015). This statistic underscores the need of an innovative approach to FM in the NLDS retail facility with (Lin et al., 2022) recommending the deployment of BIM-based Facility Management (BIM-FM).

Maintenance and Energy Management are some of the numerous areas of research in BIM-FM. (Manchana, 2022) investigates the application of ML and AI in facility management buttressing the potential of these technologies in supporting sustainable FM through monitoring and optimization of facility systems. Accordingly, Cheng et al. (2020) develops a ML-dependent framework for predictive maintenance and Kazado et al. (2019) focuses on the monitoring of indoor environment in real-time for optimization based on the operational state of the facility.

3.1 BIM-based Predictive Maintenance of Facilities

Maintenance is integral aspect of FM given its substantial share of annual operational cost Cheng et al. (2020), necessitating effective maintenance strategies in the operation of the NLDS retail facility capable of minimizing this cost with potentials of extending the service life of the facility's components. Predictive maintenance (PM) is a type of condition-based preventive maintenance that utilizes condition-monitoring of selected performance indicators such as the motor current draw rate of the HVAC system in the retail facility (Atkin & Brooks, 2021) to enable proactive maintenance and repair activities aimed at reducing the probability of failures (Global Forum on Maintenance and Asset Management, 2021). It focuses on identifying early signs of failure and gradual dilapidation by analysing trends in component conditions using historical data (Cheng et al., 2020).

Hence, BIM-based PM (BPM) in this context can be conceptualized as an advanced, proactive maintenance strategy that harnesses the comprehensive data visualization, and analytical capabilities of BIM, Internet-of-Things (IoT) sensors, and Machine Learning (ML) algorithms to anticipate, schedule and optimize maintenance activities throughout a building's life cycle. This integration provides a holistic view of risk factors, system relationship, spatial dependencies and supports pre-emptive interventions to avoid costly downtime, extending the lifecycle of built assets (Eskandari & Noorzai, 2021). Additionally, it facilitates BIModel visualization of assets for effective planning and prioritization of maintenance task based on asset conditions (Patacas et al., 2020). According to Gao and Pishdad-Bozorgi (2019), BIM-based preventive maintenance enables data driven decision-making which enhances maintenance efficiency, enabling the NLDS retail facility to operate proactively rather than reactively.

According to Cheng et al. (2020), BPM requires separate layers for information management and application, with the information layer integrating multiple sources of information and the application layer encompassing modules for monitoring, assessment, prediction and notification. Summarily, he proposes a framework for the implementation of BPM.

Equipping the escalators in the NLDS retail facility with IoT enabled accelerometers, strain gauges, acoustics and temperature sensors can provide real-time data on vibration levels, stress levels, over-heating and unusual acoustic patterns. Advanced predictive modelling and anomaly detection algorithms process this data, identifying deviations from optimal conditions. This insight can be integrated into a digital twin of the escalator in the BIM model (see Figure 7), thus improving the understanding of the physical layout, accessibility and maintenance challenges, enhancing resource allocation and optimizing maintenance strategies.

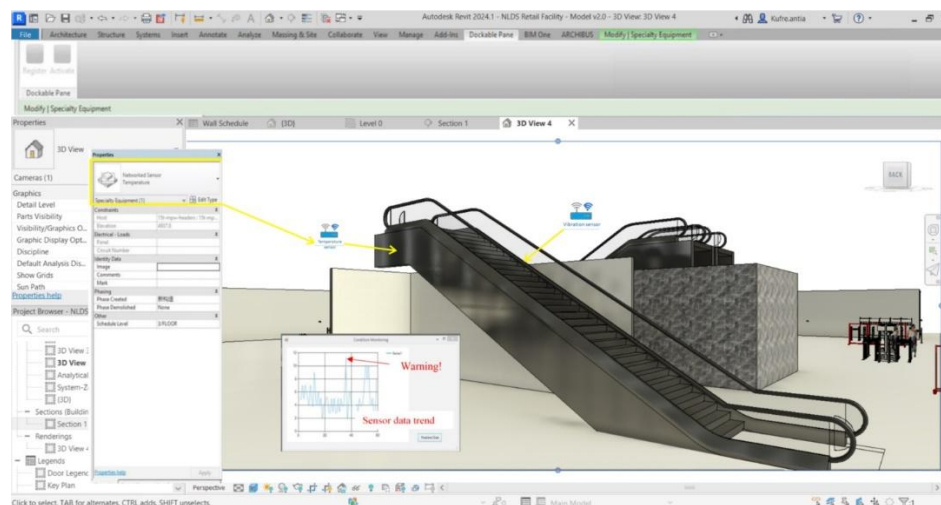


Figure 7: NLDS Retail Facility Escalator

BPM offers several significant benefits as it aids in addressing emerging issues before they cause system failures, thus extending the life span of building components, negating the cost of replacement and repairs (Atkin & Brooks, 2021). Continuous condition-monitoring enabled by IoT sensors allows for maintenance to be scheduled only when necessary, reducing

the need for operational interruption in facilities, which minimizes labour and material cost while improving operational efficiency (Cheng et al., 2020). Eskandari and Noorzai (2021) add that with real-time insight into asset conditions, facility managers can prioritize maintenance actions based on criticality, which prevents shutdowns and resource waste. Predictive maintenance contributes to sustainability, aligning with energy efficiency goals as well-maintained equipment operate effectively and consumes less energy, contributing to reduced carbon emissions (Gao & Pishdad-Bozorgi, 2019)

The need for information integration in BPM exposes the process to challenges due to quality and interoperability of data. Chan and Hsu (2019) in Manchana (2022), highlights that the effectiveness of ML and AI models in predictive maintenance is significantly influenced by the quality and accessibility of data, noting that data might be fragmented inconsistent or challenges to gather due to interoperability of multiple technological systems thus posing a risk to accuracy and dependability of predictive models. Additionally, despite the existence of standards such as the ISO 13374 and 13381-1 for condition monitoring, Manchana (2022) cites the lack of standardized frameworks for BIM and facility management integration, noting that FM operations depend on legacy systems which poses a challenge to the integration of multiple technologies such as BIM, AI and ML due to compatibility issues without significant repercussions to cost and time. ABILux (2024) highlights the resistance of FM teams to adopting new technology and processes and recommends efficient change management process and training programs to meet these challenges.

Existing challenges in the integration of FM systems with supporting technologies necessitates further research on standardization of multiphase technological integrations to guide the deployment advanced technologies, explore solutions to identified challenges and assess long-term impacts.

3.2 BIM for Energy Management

Energy management is critical to modern facility management operations as it promotes responsible use of resources and addresses environmental concerns associated with greenhouse gas emission, which directly contributes to a facility's environmental footprint, whilst improving the long-term cost efficiency of the NLDB facility. According to Hadjioannou (2023), the built environment represents 35% and 38% of global energy and Co² consumption. Liu et al. (2020) adds that more energy is consumed in the operational and maintenance phase of a facility as compared with design and construction. This underscores the urgency of transitioning towards energy efficient practices to align with global climate objectives. He considers the fragmented nature of energy data across various traditional energy systems as a hindrance to efficient energy management due to the lack of a unified platform for seamless data analysis. BIM addresses this problem by providing digital platforms that integrates complex building data for energy monitoring (Gao & Pishdad-Bozorgi, 2019). According to Khan (2022), BIM improves energy efficiency of existing buildings through enhanced visualization of real-time energy data for optimization.

Valinejadshoubi et al. (2022) research on the integration of BIM in FM focusing on the integration of IoT sensors to alleviate challenges associated with lack of visual information and data integration platform. He considers BIM as a useful tool, acting as a data repository and a visual model for representing FM information in a user-friendly manner. Accordingly, Kazado et al. (2019) presents a novel methodology for integrating energy performance sensor data in BIM for visualisation of temperature and CO² readings using Navisworks. This approach involves the integration of a 3D Models and Sensor data for continuous surveillance of temperature and CO² levels. These systems can monitor and analyse building performance in real-time, reveal temperature fluctuations during occupied and unoccupied hours together with

CO² levels. It enables the identification of patterns and anomalies for proactive adjustments to HVAC systems and other building operations in FM.

BIM-FM systems can integrate with Building management systems and energy monitoring tools to allow continuous tracking and optimization of energy use by managing data on heating, cooling and lighting demand, aligning these with real-time occupancy and environmental data for optimization. It offers an opportunity for improvement through the utilization of AI-driven automation to predict and effectively manage smart HVAC systems at different periods depending on a facilities usage condition, real-time feedback of customers from sentiment analysis and preference, thus boosting the attainment of facility, cooperate and sustainability goals and employee and customer satisfaction (Manchana, 2022). In addition to improving energy efficiency, BIM enhances sustainability efforts by optimizing resource management and minimizing carbon footprint of a facility. BIM based tools for sustainability analysis evaluate environmental performance through life cycle assessments, there-by lowering emissions linked to building operations.

Despite the benefits of BIM-FM systems in energy management, the need for integration across multiple platforms births challenges due to compatibility and data interoperability. According to Khan (2022), lack of compatibility accounts for two-thirds of interoperability expenditures. Along with compatibility, thus interoperability concerns, Liu et al. (2020) highlights the need for early introduction of facility managers in a projects life cycle to facilitate the clear definition of FM information requirements. Additionally, he recognises the need for clearly defined processes to guide BIM and FM integration. The standardization of processes for energy management systems can positively influence the deployment of BIM in FM by streamlining technology and enhancing interoperability.

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